Allie (Minh-Anh) Nguyen

@ allienguyen@berkeley.edu | +1 510 255 5038 | linkedin.com/in/alliemnguyen

Education

University of California, Berkeley — Expected: Dec 2027
Ph.D. Candidate, Environmental Engineering, GPA: 4.0/4.0 — *Berkeley, CA*

- Dissertation: "Microbial Quality in Urban Water Systems: Virus Persistence, Neighborhood Surveillance, and Direct Potable Reuse"
- Advisor: Professor Kara Nelson
- Minors: Biostatistics, Epidemiology

University of California, Berkeley — Aug 2022 – Dec 2023
Master of Science, Environmental Engineering, GPA: 4.0/4.0 — *Berkeley, CA*

University of Minnesota - Twin Cities — Aug 2018 – May 2022
Bachelor of Environmental Engineering, with Distinction, GPA: 3.81/4.0 — *Minneapolis, MN*

Research Interest

Wastewater-based surveillance, water-borne pathogens, water and wastewater treatment processes, water reuse, urban microbial water safety, applications of water and wastewater treatment on public health.

Research Experience

Graduate Student Researcher — Aug 2022 – Present
University of California, Berkeley*, Department of Civil and Environmental Engineering* — *Berkeley, CA*

- **Persistence of Viral Nucleic Acids in Wastewater:** Design and execute novel experiments to study how surfactants influence the decay of viral nucleic acid signals in wastewater across 8 human respiratory viruses.
- **Sub-sewershed Sampling for Wastewater Surveillance:** Conduct extensive data analysis on San Francisco sub-sewersheds' COVID-19 wastewater and case datasets to evaluate how neighborhood-scale sampling can improve health equity in wastewater-based surveillance.
- **Supplementing Conventional Drinking Water with Direct Potable Reuse Water:** Develop protocols, lead experiments, and contribute in building experimental models to investigate the microbial quality of advanced treated water blended with conventional surface water in pilot distribution systems.
- Mentor 4 master's students on microbial experimental techniques.

Undergraduate Research Assistant — Jan 2021 – Aug 2022
University of Minnesota Twin Cities,
Department of Civil, Environmental, and Geo- Engineering — *Minneapolis, MN*

Project: ***Biochar for Nutrient Removal from Municipal Wastewater*** — Sep 2021 – Aug 2022

- Advisors: Professor Sebastian Behrens, Dr. Mariah Dorner.
- Operated, prepared synthetic wastewater, and collected samples for 6 biochar-amended sequencing batch reactors.
- Analyzed nitrate, nitrite, and ammonium using segmented flow analysis and UV-Vis to characterize initial sorption isotherms of ammonia to biochars and DAX-8 materials in water.

Project: ***Hydrologic & Hydraulic Characteristics of Effective Sea Lamprey Barriers*** — Jan 2022 – May 2022

- Advisors: Professor Miki Hondzo, Dr. Daniel P. Zielinski.
- Conducted literature review on the efficacy of current sea lamprey barrier technologies.
- Designed and performed fluid mechanics experiments with model fixed-crest barriers in hydraulic flumes.
- Analyzed the time-averaged and root-mean-square velocities of the water flow due to barriers' impact, lamprey detachment forces, and turbulent kinetic energy from the experiment results.

Project: ***Biomethane Generation Potential of Domestic Municipal Waste*** — Jan 2021 – Nov 2021

- Collected samples and evaluate composition of organic municipal waste in Hanoi, Vietnam.
- Performed bench-scale biochemical methane potential tests and measured anaerobic digestion capability of sample organic waste resembling real-life waste composition.

Publications

- **Nguyen, M-A.**; Ung, C.T.L.; Dao, A.D.; Dinh, C.L.; Nguyen, V-A.. Wastewater reuse in Vietnam: potential, challenges, and strategic solutions. *Water Reuse* **2026**; *16 (2)*: 379–399. https://doi.org/10.2166/wrd.2026.152.
- Wang, A.L.; Jiang, M.; **Nguyen, A.**; Kane, S.R.; Borucki, M.K.; Kantor, R.S.; Nelson, K.L.. Benchmarking Concentration and Extraction Methods for Wastewater-Based Surveillance of Eight Human Respiratory Viruses: Implications for Rapid Application to Novel Pathogens. *Environmental Science & Technology* **2025**, *59* (36), 19107–19118. https://doi.org/10.1021/acs.est.4c13635.

Presentations

- **Nguyen, A.**; Nelson, K.. *Persistence of viral nucleic acid signals in the presence of surfactants in wastewater.* Poster presented at the West Coast Wastewater Surveillance for Public Health Conference, Sacramento, California, USA, July 21-22, 2026.
- **Nguyen, A.**; Nelson, K.. *Persistence of viral nucleic acid signals in the presence of surfactants in wastewater.* Poster presented at the Gordon Research Conference "Microbes in Built Environments", Waterville Valley, New Hampshire, USA, June 21-27, 2025.
- **Nguyen, A.**; Nguyen, V.A.; Ha, D.M.; Duong, T.H.. *Pretreatment of Organic Waste and Sewage Sludge for Biogas and Methane Production Improvement in Anaerobic Digestion in Hanoi, Vietnam.* Poster presented at the IWA World Water Congress & Exhibition, Toronto, Canada, August 11-15, 2024.
- **Nguyen, A.**; Tschirner, U.. *Determination of Biomethane Generation Potential in the Domestic Municipal Waste Flow of Hanoi, Vietnam".* Poster presented at the University of Minnesota Fall 2021 Undergraduate Research Symposium, virtual (Minneapolis, MN, USA), November 27, 2021.

Teaching and Mentorship Experience

Graduate Student Instructor — Jan 2025 – Present
***University of California, Berkeley**, Department of Civil and Environmental Engineering* — *Berkeley, CA*
Course: Introduction to Environmental Engineering (CE 115) — *Aug 2026 – Present*

- Supervising lecturer: Professor Baoxia Mi

Course: Environmental Biological Processes (CE 211B) — *Jan 2025 – May 2025*

- Supervising lecturer: Professor Amy Pickering
- Collaborated with course instructor to revise assignments and develop grading rubrics.
- Graded weekly homework assignments and exams of 35 graduate students.
- Held weekly office hours and led exam review sessions to reinforce key concepts, provide example problems, and enhance student comprehension.

SURF Program Graduate Student Mentor — Jan 2023 – Aug 2023
***University of California, Berkeley**, Office of Undergraduate Research and Scholarships* — *Berkeley, CA*

- Led draft proposal workshops and weekly office hours for applicants of the Summer Undergraduate Research Fellows (SURF) program.
- Led two workshops on research skills with an attendance of 50 SURF fellows.
- Organized and facilitated the SURF program's orientation, weekly small group meetings, and SURF Research Symposium.

Undergraduate Student Instructor — Jan 2022 – May 2022
***University of Minnesota Twin Cities**,*
Department of Civil, Environmental, and Geo- Engineering — *Minneapolis, MN*
Courses: Computer Applications (CEGE 3101); Introduction to Environmental Engineering (CEGE 3501)

- Graded assignments for 60 upper-division undergraduate students.
- Collaborated with course instructor to develop grading rubrics.

***University of Minnesota Twin Cities**, College of Science and Engineering* — *Minneapolis, MN*
Course: First-Year Experience (CSE 1001)

- Prepared teaching materials, held lectures, and managed class logistics for 30 first-year undergraduate students.

SMART Peer Tutor Mentor — Sep 2019 – May 2022
***University of Minnesota Twin Cities**, SMART Learning Commons* — *Minneapolis, MN*

- Assisted over 100 students' learning inquiries in Math and Chemistry.
- Observed and evaluated 12 SMART peer tutors with 1-on-1 consultation sessions and written reports.
- Led a 60-minute staff meeting every semester for additional training of peer tutors.

Honors and Awards

- University of California Berkeley, The Trussell Fellowship in Environmental Engineering 2022–2023
- University of Minnesota, College of Science and Engineering Dean's List 2018-2021
- University of Minnesota, Department of Civil, Environmental and Geo- Engineering James and Sharon Weinel Chi Epsilon Scholarship 2021–2022
- University of Minnesota, International Student and Scholar Services Culture Corps Program Scholarship 2021
- University of Minnesota, Undergraduate Research Opportunities Program Scholarship 2021
- University of Minnesota, Department of Civil, Environmental and Geo- Engineering Adolph Sommerfeld Scholarship 2020
- University of Minnesota, Global Excellence Scholarship 2018–2022

Additional Work Experience

International Student Consultant May 2021 – Aug 2021

University of Minnesota Twin Cities, *International Student and Scholar Services* *Minneapolis, MN*

- Assisted 80 incoming international students during orientation and before their US arrival.
- Provided logistical support to the CSE Academic Advising office in facilitating activities and preparing for orientation.

Environmental Engineering & Management Intern May 2019 – Jan 2021

Hanoi University of Civil Engineering, *Institute of Environmental Science and Engineering* *Hanoi, Vietnam*

- Participated in preparation of five environmental engineering research papers, manuals, and reports.
- Contributed to environmental management consultancy contracts of the Institute of Environmental Science and Engineering.
- Conducted site visits to wastewater treatment plants, clients' factories, and contaminated sites, and collected field surveys of municipal waste management.

Leadership, Service, and Community Involvement

Beyond Academia, UC Berkeley Apr 2023 – Jun 2026

Logistics Co-director May 2024 – Jun 2026

- Oversee logistics for all events and operations, including the Beyond Academia conference with 650+ attendees from over 20 countries.
- Serve as the main contact for external partners, including UC Berkeley Graduate Division, Graduate Professional Development, and Berkeley Career Engagement.
- Lead meetings with members and co-directors, setting event logistics frameworks.

Finance Co-director Jan 2024 – May 2024

- Managed a $70,000+ for Beyond Academia, a non-profit student-led organization.
- Co-directed and managed a team of 10+ graduate student volunteers.
- Organized, recruited guest PhD speakers, and moderated panels for the annual Beyond Academia conference (1,300+ attendees).

Graduate Women of Engineering (GWE), UC Berkeley Sep 2022 – Jun 2026

Co-President Jun 2024 – Jun 2026

- Lead and oversee operations of GWE, including officer meetings, communications, elections, and administration compliance.
- Serve as primary liaison with faculty, campus offices, and partner student organizations; manage official inbox and external communications.
- Coordinate strategic initiatives and events, including newsletters, inter-organization collaborations, and external professional opportunities.

Career and Professional Development Committee Head Jun 2023 – Jun 2024

- Organized the annual Career Panel, Speaker Series, and other professional development events for women-identifying engineering graduate students at UC Berkeley.

Other Professional and Campuswide Service, UC Berkeley — 2024 – Present

- Lab Manager and Lab Safety Coordinator, Nelson Research Group — Sep 2024 – Present
- Core member, Climate Equity and Environmental Justice Faculty Search Committee — Dec 2025 – Feb 2026
- Core member, CEE Environmental Engineering Graduate Admission Committee — Jan 2026 – Feb 2026
- Co-coordinator, CEE Environmental Engineering Seminar Series — Jan 2024 – May 2024

Activities at the University of Minnesota – Twin Cities

- American Society of Civil Engineers — Historian (2021-22)
- CSE International Ambassadors — President (2021-22), Vice President (2020-21), Secretary (2019-20)
- Chi Epsilon Honors Society — Vice President (2021-22)
- Design U — Research and Strategy Team Member (2021)
- Welcome Week Program — Welcome Week Leader (2020)

Skills

- **Laboratory:** Digital PCR, DNA/RNA extraction from water and wastewater, UV–Vis spectroscopy, cell culture, sequencing batch reactor operation, chemical and microbial analysis of water quality.
- **Software:** R/RStudio, LaTeX, MATLAB, Microsoft Office.
- **Languages:** Vietnamese (native speaker proficiency), English (professional proficiency).
- **Certifications:** UC Berkeley Graduate Certificate in Applied Data Science (in progress), Google Data Analytics Professional Certificate (via Coursera), Master Certified Tutor (CRLA Level 3).